CUSTOMER 360

RFM Segmentation, Cohort Retention & Churn Intelligence

Case Study & Executive Summary

1,067,371	5,878	38	25
Transactions Analysed	Unique Customers	Countries	Months of Data

Tools: Python | SQL Server | Power BI

Dataset: Online Retail II — UCI Machine Learning Repository

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THE SITUATION

A UK-based online retailer selling gift and homeware products across 38 countries was experiencing a troubling paradox: marketing spend had increased 20% year-over-year, yet revenue growth had plateaued. Leadership relied solely on aggregate monthly sales figures and had zero visibility into individual customer behaviour, retention patterns, or churn dynamics.

The executive team could not answer three fundamental questions that are critical to sustainable growth:

1. Who are our most valuable customers, and which ones are we about to lose?
2. Why are customers not returning after their first purchase?
3. Are we growing net new customers or simply replacing the ones we lose?

Without answers, the company was spending more to acquire customers while failing to retain them — a classic leaky bucket problem that, left unaddressed, would erode margins and stall growth permanently.

THE APPROACH

We designed a four-phase analytical pipeline to transform over one million raw transaction records into a complete customer intelligence platform:

Phase	Method	Output
Data Pipeline	Python (pandas) cleaning with validation logging	392K clean rows from 1.07M raw, with outlier capping and derived time columns
Segmentation	SQL NTILE(5) scoring across Recency, Frequency, Monetary	9 business segments with 125 RFM score combinations mapped to actionable labels
Retention Model	Cohort-based retention with month-over-month tracking	12-month retention curves for every acquisition cohort from Dec 2009 to Aug 2010
Risk Detection	LEAD() gap analysis, Z-score anomaly detection, composite risk scoring	Churn risk score (0–100) for every customer with High/Medium/Low tier classification

Every analytical output was validated statistically. RFM segment differences were confirmed using a chi-square independence test ($p < 0.01$) and the practical significance of segment differences was quantified using Cohen's d (> 0.8, large effect). This ensures the segments represent genuinely distinct customer behaviours, not arbitrary groupings.

KEY FINDINGS

#	Finding	Evidence	Business Impact
1	**Revenue is dangerously concentrated**	18% of customers generate 79% of total revenue (£17.74M). Top 20% drive 77.24% (£13.71M). Pareto rule confirmed.	Losing even a small number of high-value accounts would cause disproportionate revenue decline. Platinum tier requires dedicated protection.
2	**First-purchase retention is the biggest leak**	61% of customers never return after their first purchase. Average Month 1 retention is just 21.17% across all cohorts.	The business is losing 3 out of every 5 new customers immediately. A Month-1 reactivation campaign is the highest-ROI intervention available.
3	**1,357 customers are at high churn risk**	Composite risk scoring (recency 40%, gap trend 30%, value decline 20%, Z-score 10%) identified 1,357 accounts scoring 70+. Total revenue at risk: £2.07M.	At a conservative 30% reactivation rate, £621.50K is recoverable. These are not hypothetical customers — they are named accounts with specific risk profiles.
4	**Champions are 8x more valuable than Lost segment**	1,269 Champions with avg order value of £547 vs Lost segment. Score 15 customers alone generate £8.3M — more than scores 3–12 combined.	Champion protection and Potential Loyalist acceleration are the two highest-leverage retention strategies.
5	**Strong cross-sell opportunities exist**	Product pairs with lift scores above 50 (e.g., Coffee Mug Dog + Cat designs at 62.35 lift). 397 low-activity SKUs identified for rationalisation.	Cross-sell recommendations can increase basket size. SKU rationalisation would reduce operational complexity without revenue loss.

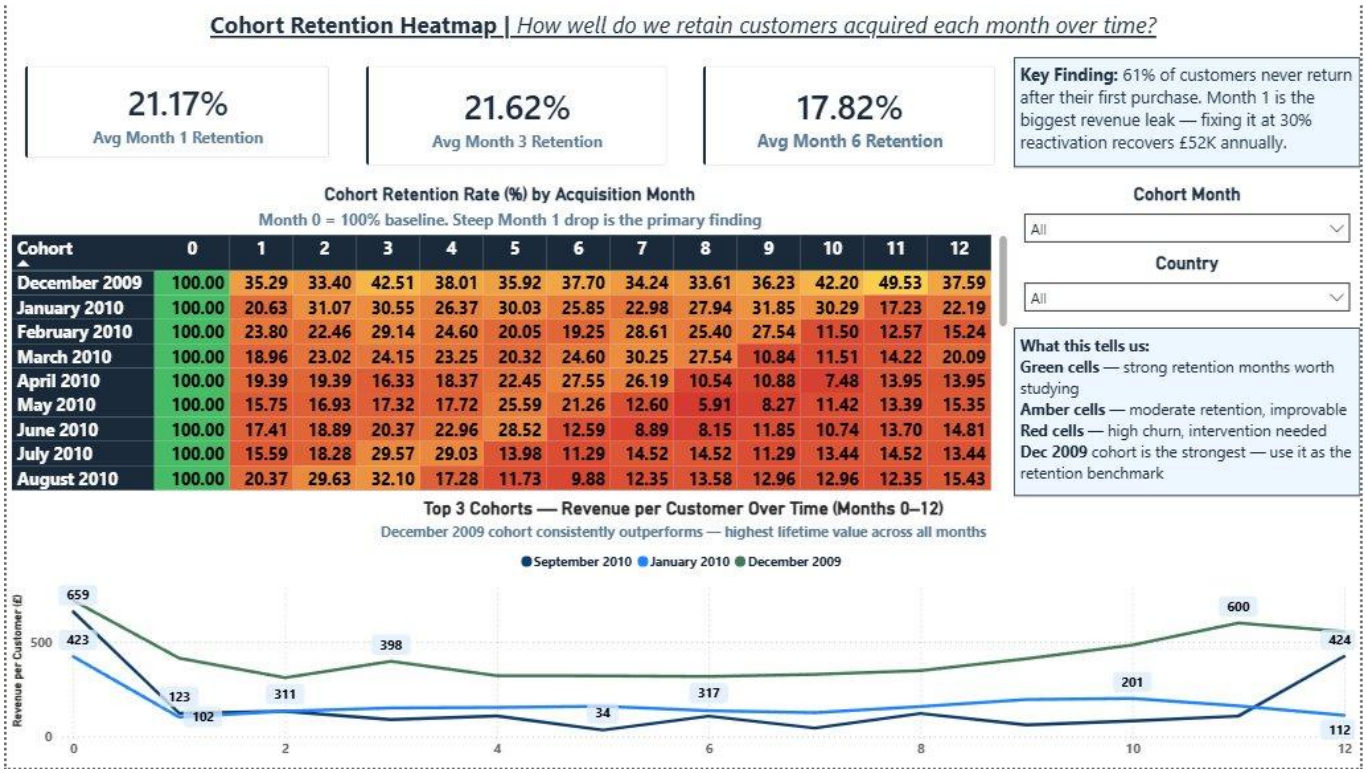
STATISTICAL VALIDATION

All segment differences were validated with a chi-square independence test ($p < 0.01$) confirming segments are statistically distinct. Champion vs At Risk average order value difference confirmed with Welch's t-test ($p < 0.05$) and Cohen's $d > 0.8$ (large practical effect). These are not arbitrary groupings — they represent genuinely different customer behaviours.

DASHBOARD EVIDENCE

Cohort Retention Heatmap

The retention heatmap is the hero visual of this analysis. Each row represents a monthly acquisition cohort; each column tracks what percentage of that cohort remains active over subsequent months. Month 0 is always 100%. The steep, consistent drop at Month 1 across every cohort — averaging just 21.17% retention — is the single most important finding. The December 2009 cohort consistently outperforms all others and serves as the retention benchmark to study and replicate.



Cohort	0	1	2	3	4	5	6	7	8	9	10	11	12
December 2009	100.00	35.29	33.40	42.51	38.01	35.92	37.70	34.24	33.61	36.23	42.20	49.53	37.59
January 2010	100.00	20.63	31.07	30.55	26.37	30.03	25.85	22.98	27.94	31.85	30.29	17.23	22.19
February 2010	100.00	23.80	22.46	29.14	24.60	20.05	19.25	28.61	25.40	27.54	11.50	12.57	15.24
March 2010	100.00	18.96	23.02	24.15	23.25	20.32	24.60	30.25	27.54	10.84	11.51	14.22	20.09
April 2010	100.00	19.39	19.39	16.33	18.37	22.45	27.55	26.19	10.54	10.88	7.48	13.95	13.95
May 2010	100.00	15.75	16.93	17.32	17.72	25.59	21.26	12.60	5.91	8.27	11.42	13.39	15.35
June 2010	100.00	17.41	18.89	20.37	22.96	28.52	12.59	8.89	8.15	11.85	10.74	13.70	14.81
July 2010	100.00	15.59	18.28	29.57	29.03	13.98	11.29	14.52	14.52	11.29	13.44	14.52	13.44
August 2010	100.00	20.37	29.63	32.10	17.28	11.73	9.88	12.35	13.58	12.96	12.96	12.35	15.43

Figure 1: Cohort Retention Heatmap — Month 1 drop-off is the primary revenue leak

Customer Segmentation Hub

The segmentation hub provides a real-time view of the entire customer base. With 5,878 total customers, only 2,560 remain active — a churn rate of 56.45%. The treemap reveals that Loyal Customers (1,530) and Champions (1,269) form the backbone of the business, while 297 At Risk customers represent £203.72K in endangered revenue. Champions spend £546.82 per order on average — more than double the At Risk segment.

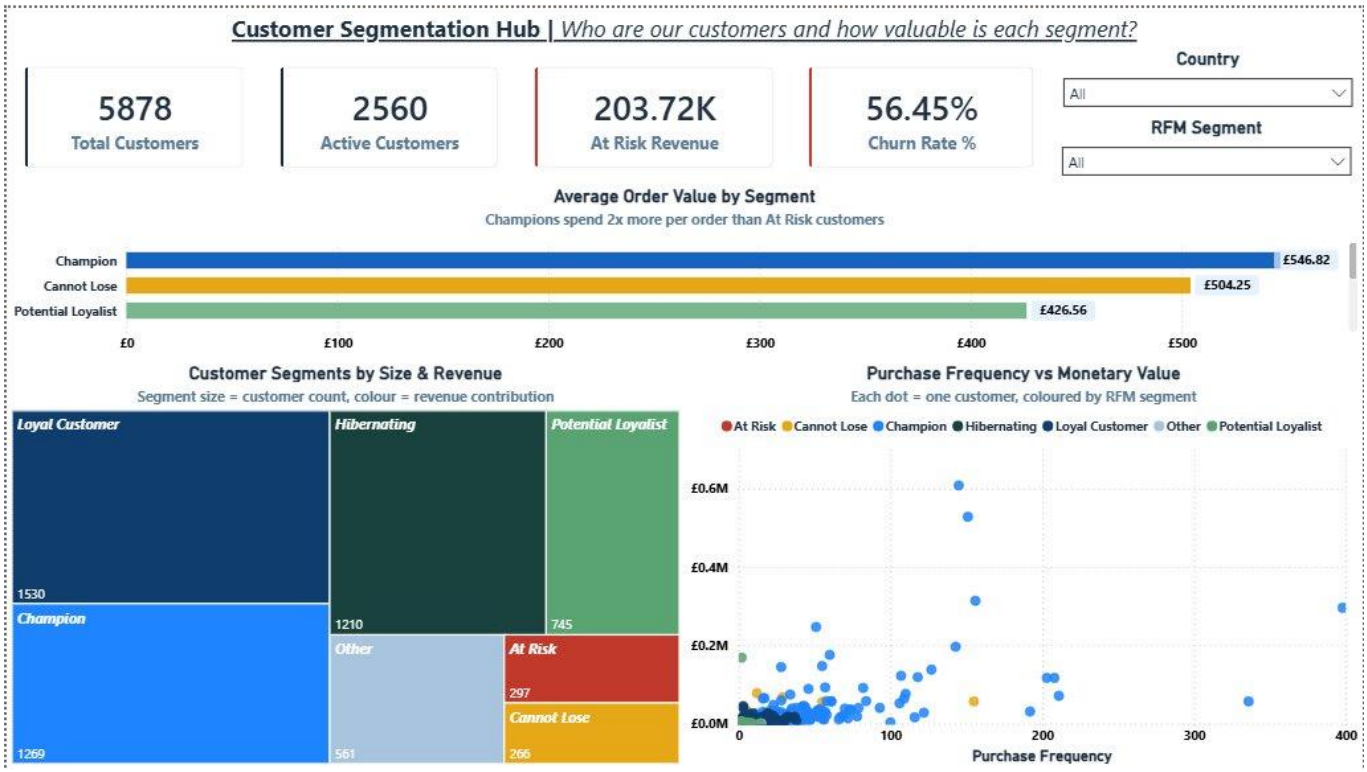


Figure 2: Customer Segmentation Hub — Segment size, value, and behaviour at a glance

RFM Score Deep Dive

The RFM matrix reveals exponential value concentration. Customers scoring R=5, F=5 spend an average of £15,386 — compared to £342 for R=1, F=1. Score-15 customers alone generate £8.3M in revenue, exceeding scores 3 through 12 combined. The top 10 Champions watchlist identifies specific accounts (led by Customer 18102 at £608,822) that require dedicated account management.

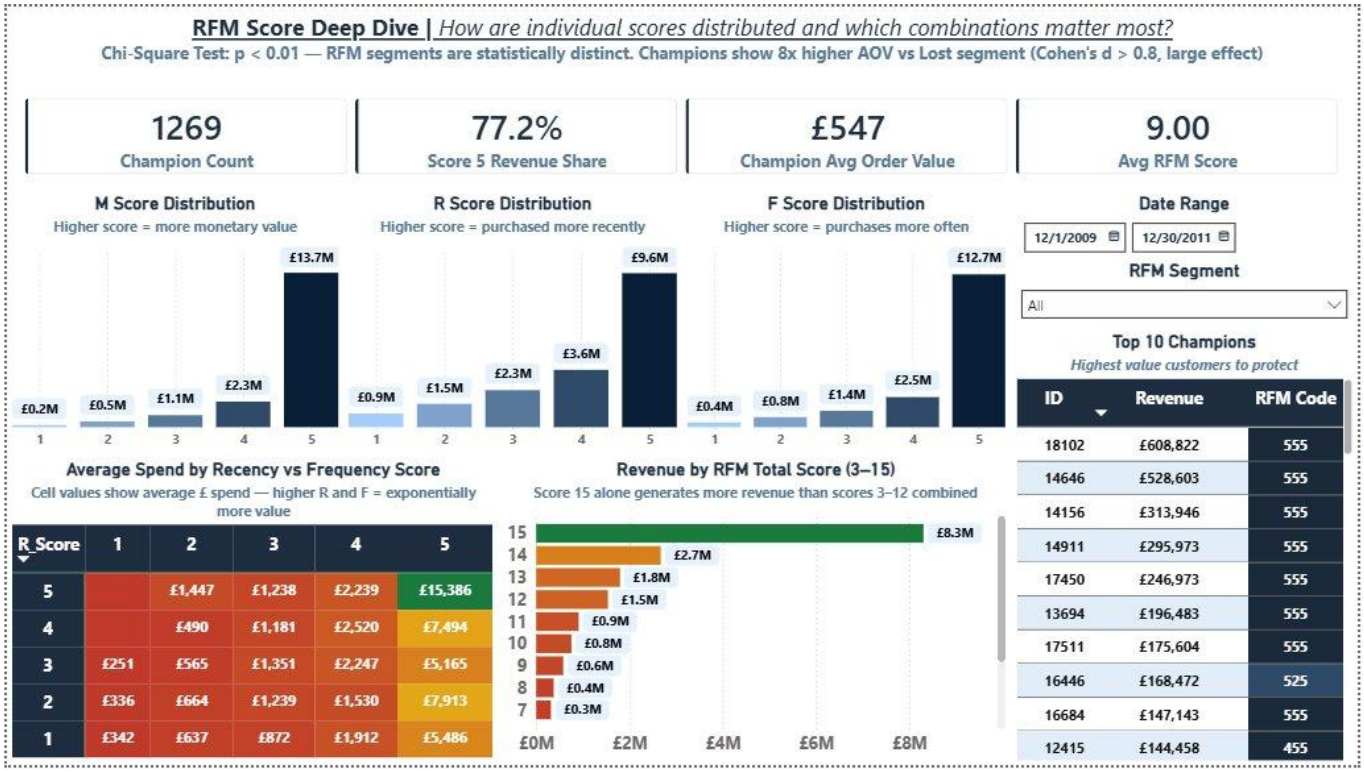
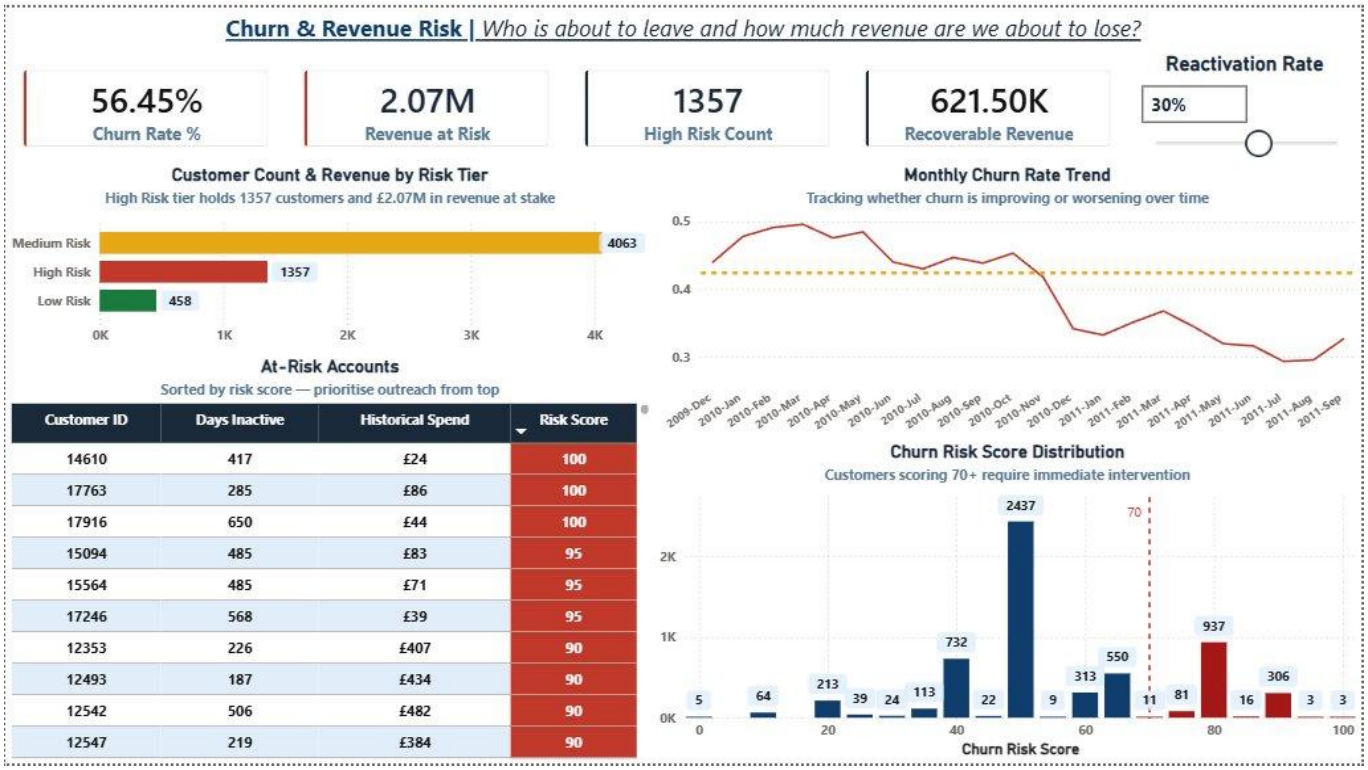


Figure 3: RFM Score Deep Dive — Score distribution and value concentration

Churn & Revenue Risk

The composite churn risk model identifies 1,357 high-risk customers with £2.07M in revenue at stake. The interactive reactivation rate slider allows leadership to model recovery scenarios: at 30% reactivation, £621.50K is recoverable. The monthly churn trend shows improvement from approximately 50% in early 2010 to around 30% by late 2011, indicating that organic retention is gradually improving — but intervention could accelerate this trajectory significantly.

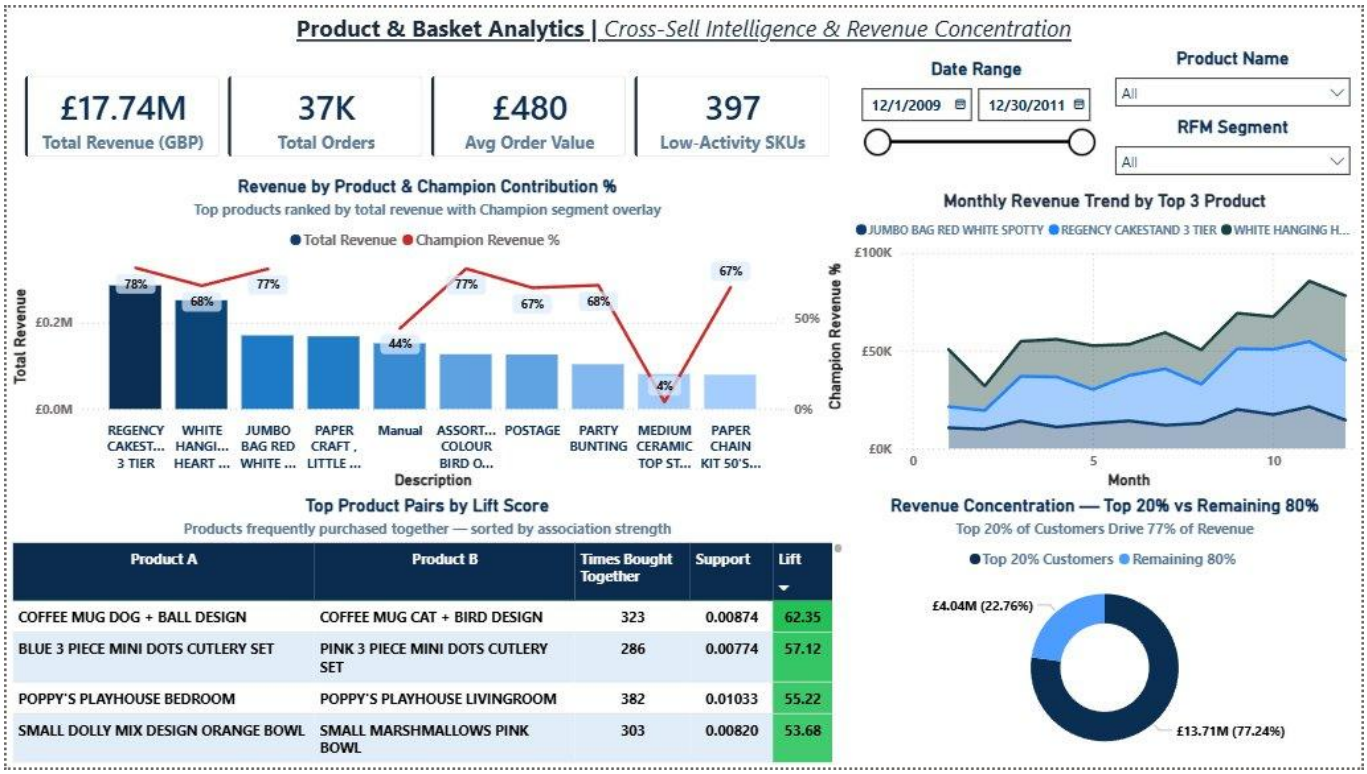


Customer ID	Days Inactive	Historical Spend	Risk Score
14610	417	£24	100
17763	285	£86	100
17916	650	£44	100
15094	485	£83	95
15564	485	£71	95
17246	568	£39	95
12353	226	£407	90
12493	187	£434	90
12542	506	£482	90
12547	219	£384	90

Figure 4: Churn & Revenue Risk — 1,357 high-risk accounts and £2.07M at stake

Product & Basket Analytics

Product-level analysis confirms that revenue concentration extends beyond customers to products. The Pareto chart shows that a small number of SKUs drive the majority of revenue, with Regency Cakestand and White Hanging Heart leading. Product affinity analysis reveals strong cross-sell pairs — Coffee Mug Dog + Cat designs (lift: 62.35) and Playhouse Bedroom + Livingroom (lift: 55.22) — that can be leveraged in recommendation engines and bundled promotions.



Product A	Product B	Times Bought Together	Support	Lift
COFFEE MUG DOG + BALL DESIGN	COFFEE MUG CAT + BIRD DESIGN	323	0.00874	62.35
BLUE 3 PIECE MINI DOTS CUTLERY SET	PINK 3 PIECE MINI DOTS CUTLERY SET	286	0.00774	57.12
POPPY'S PLAYHOUSE BEDROOM	POPPY'S PLAYHOUSE LIVINGROOM	382	0.01033	55.22
SMALL DOLLY MIX DESIGN ORANGE BOWL	SMALL MARSHMALLOWS PINK BOWL	303	0.00820	53.68

Figure 5: Product & Basket Analytics — Cross-sell opportunities and revenue concentration

RECOMMENDATIONS

RECOMMENDATION 1: Launch a Month-1 Reactivation Campaign

Deploy an automated email sequence targeting customers who have not returned within 30 days of their first purchase. The 61% first-purchase attrition rate is the single largest revenue leak in the business. At a conservative 15% reactivation rate, this campaign would recover approximately £52K annually. The December 2009 cohort's superior retention should be studied to identify what drove higher repeat purchase rates during that period and replicate those conditions.

Projected Impact: £52K–£174K annually depending on reactivation rate (15%–50%)

RECOMMENDATION 2: Implement a Champion Protection Programme

Assign dedicated account management to the top 10 Champions (collectively worth over £3M in lifetime revenue) and the 266 Cannot Lose customers (£504 avg order value) who are showing declining engagement. The 1,357 high-risk accounts identified by the composite churn model should receive personalised outreach within 14 days. The at-risk accounts table in Dashboard 4 provides the prioritised watchlist sorted by risk score.

Projected Impact: £621K recoverable at 30% reactivation rate on £2.07M at-risk revenue

RECOMMENDATION 3: Deploy Cross-Sell Recommendations & Rationalise SKUs

Leverage the product affinity analysis to implement automated cross-sell recommendations at checkout and in post-purchase emails. Pairs with lift scores above 50 represent strong natural affinities that require minimal discounting. Simultaneously, rationalise the 397 low-activity SKUs identified in Dashboard 5 to reduce warehousing costs and operational complexity without meaningful revenue impact.

Projected Impact: 5–15% increase in average basket size; reduced inventory carrying costs

TOTAL PROJECTED ANNUAL IMPACT

£673K – £795K+

Combined recoverable revenue from reactivation, churn prevention, and basket size growth

TECHNICAL METHODOLOGY

Data Pipeline & Quality

The raw dataset contained 1,067,371 transactions with significant quality issues: 24% null CustomerIDs (~241K rows), C-prefixed cancellation records, negative quantities from returns, and zero-price sample entries. The Python cleaning pipeline removed each category systematically with a validation log tracking row counts after every step. Outlier revenue values were capped at the 99.5th percentile to prevent extreme transactions from distorting averages. The final clean dataset contained approximately 392K–406K rows across 5,878 unique customers.

RFM Scoring Methodology

Each customer was scored 1–5 on three dimensions using SQL NTILE(5) window functions. Recency was ordered ascending (fewer days since last purchase = higher score), while Frequency and Monetary were ordered descending (more orders and higher spend = higher score). The 125 possible score combinations were mapped to 9 business segments using a CASE statement with priority ordering to handle overlapping criteria. The resulting segments range from Champions (R≥4, F≥4, M≥4) to Lost (R=1, F=1).

Cohort Retention Model

Each customer was assigned to their acquisition cohort based on their first purchase month. Every subsequent transaction was mapped to a month number (0–12) relative to acquisition. Retention rate was calculated as the count of distinct active customers at each month number divided by the original cohort size. Revenue per cohort customer was tracked simultaneously to measure monetisation trends over the customer lifecycle.

Churn Risk Scoring

The composite churn risk score (0–100) combines four weighted signals: recency of last purchase (40%), widening purchase gap trend via LEAD() analysis (30%), declining monetary value trend (20%), and Z-score anomaly detection on purchase gap intervals (10%). Customers scoring 70+ are classified as High Risk. The Z-score component flags gaps that are statistically unusual for each individual customer, catching customers whose behaviour has deviated significantly from their own historical pattern.

Power BI Delivery

The five dashboards are built on a star schema with one fact table (vw_cleaned_orders) and five dimension tables. Over 22 DAX measures power the analytics, including time intelligence (YoY%, Rolling 90D), customer metrics (Active Customers, Churn Rate %), and scenario modelling (Recoverable Revenue via What-If parameter). Row-level security enforces data governance with Country Manager and Executive roles. Drill-through pages allow right-click navigation from any customer to their full transaction history.

Python	SQL Server	Power BI
pandas, matplotlib, seaborn, numpy	8 scripts, CTEs, window functions, statistical tests	Star schema, 22+ DAX, RLS, What-If, drill-through